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Case Study in Crew Management

Independent Test Project Design Team

WorldSkills Abu Dhabi 2017 Proposed Project

AMONIC Airlines

CASE STUDY IN CREW MANAGEMENT

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INTRODUCTION | ONE

The management at AMONIC Airlines plans to disband their current standalone system of crew planning. They would like to have an integrated one, which will give them the option to fill out the flights with the required staff.

The current database and some sample data related to this section are made available. After setting up the database, and reading the instructions, you will need to provide the requested diagrams.

DESCRIPTION | TWO

The aircrew schedule for a month will need to be finalized one week before the start of each month. The way this works is that at least thirty days before, when the flights for the month are planned out, the employees will have a chance to post their preferential bidding and requests. Seven days before the start of the month, the bidding system will stop taking bids. The management will then fill out the positions required taking into account the bids (accepting or rejecting them), open positions left, and all the constraints (described below) and provide crew rosters to the staff.

The personnel required as flight crew will either have qualifications to be a pilot or a cabin member. They will be assigned to positions based on their skills (both qualification and experience).

The following are some specifics on the positions available on each flight:

- Flight deck:
  - Captain: The most skilled (the most hours clocked in cockpit) pilot (1 Person)
  - First Officer: Second in command, also a pilot (1 Person)
  - Second Officer: aka flight engineer, also a pilot (1 Person)
- Cabin positions:
  - Purser: aka Cabin Service Director who is responsible for the team, a flight crew (the most experience) (1 Person)
  - Flight attendant: Number depends on the number of seats on the aircraft:
    - One crew member for each twenty seats on economy
    - One crew member for each seven seats on business and first class
    - Minimum number of flight attendants for a flight is five

Here are some of the constraints on filling out the positions:

- Every aircraft that leaves for a destination other than Abu Dhabi (the main hub) would need to keep the same crew on schedule until it arrives back.
- On flights longer than 4 hours to any destination other than Abu Dhabi, the crew will rest for at least eight hours before their next flight. This time goes on their work schedule (is paid) but not flight time.
- Each employee cannot fly more than 50 hours a week
- There will always be one set of crew representing the largest crew needed for a flight on reserve in case of emergencies.

Technical limitations for you to consider:

- The included tables of the database given, must be included in your analysis where applicable (they have a direct part in the structure). Here is how to incorporate them into your project:
  - The tables or fields on the tables cannot be removed or altered.
  - Adding fields to the tables are permitted.
- All deliverables should be in PDF format ready to be shared and printed

DELIVERABLES | THREE

In order to have this section of the software development implemented, AMONIC Airlines is asking for technical documentation in form of diagrams. All diagrams need to be submitted in the form of pdf files. The tool of choice to draw diagrams is MS Visio.

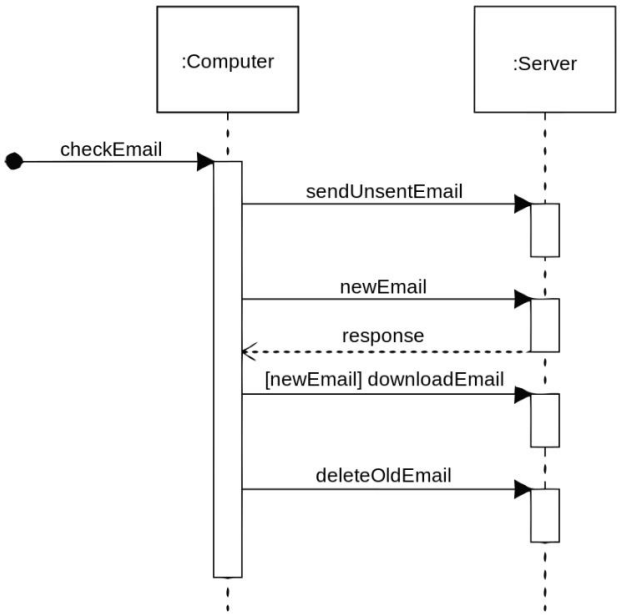
Crew Planning ERD

The requirement for this section is an overall standard Entity Relationship Diagram that includes all the required entities and their relationships. The ERDs used throughout the test project should be a good model to base your work on.

Sequence Diagram for Preferential Bidding

As specified under the description section of this document, the personnel can bid on how they would like to work each month with the new integrated system. Please provide a UML sequence diagram (event diagram) of how this is going to be implemented in order to let the human resources department to review it.

A sequence diagram is an interaction diagram that shows how objects operate with one another and in what order. It is a construct of a message sequence chart.

A sequence diagram shows object interactions arranged in time sequence. It depicts the objects and classes involved in the scenario and the sequence of messages exchanged between the objects needed to carry out the functionality of the scenario. Sequence diagrams are typically associated with use case realizations in the Logical View of the system under development.

The following is a sample for you to model your diagram on: